Network Layer Architecture & Router Design

Coverage: Data Plane (Forwarding) vs Control Plane (Routing), Datagram Networks vs Virtual Circuits, Router Fabrics & Buffering

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1. Data Plane vs Control Plane

- Data Plane: Local nanosecond hardware packet forwarding (ASIC/TCAM).
- Control Plane: Network-wide software routing determining end-to-end paths (OSPF, BGP, SDN).

2. Datagram vs Virtual Circuit

- Datagram: Stateless routers, independent per-packet routing. Virtual Circuit: Connection setup, stateful switches.

■ KTU University High-Yield Exam Questions

- Q: Compare Datagram networks and Virtual Circuit networks. (8 Marks)